

SID4: 1421
PORT_BASE: 8521
PREFIX: s1421
SEED: 1421
VERIFY_SEED: 261421
DOMAIN_ID: 5
Hardware: Apple M2 Pro, 16GB
Local model used:
Tagged commit hash:

Part 1 – I HTML

1.

```
<title>HW1-Eric Zhao</title>
```

2.

<h1 id="page-heading">Local Restaurant Inspection</h1>

ers/eric/data260/hw1/HW1-%7BERic%20Zhao%7D.html

Local Restaurant Inspection

3.

DOMAIN_SCHEMA.md x

DOMAIN_SCHEMA.md Preview

Domain Schema: Local Restaurant Inspections

Fields

Field	Type	Required	Purpose
restaurantID	Text	Yes	Primary field uniquely identifying the inspected restaurant.
restaurantName	Text	Yes	Secondary field giving the restaurant's name.
Email	Email	Yes	Email address of the submitter.
Comments	Text area	Yes	Detailed content describing observations and inspection findings.
Result	Category	Yes	Inspection result selected from the four category values below.
termsAccepted	Boolean	Yes	Records agreement to the terms and conditions.
submissionDate	ISO 8601 date-time	Generated	Date and time added by JavaScript after successful validation.

Category Values

The Result field accepts exactly one of these values:

1. Excellent Pass

2. Pass

3. Needs Reinspection

4. Fail

4.

a.

```
<form>
  <div class="form-field">
    <label for="restaurantID">Restaurant ID</label>
    <input
      id="restaurantID"
      name="restaurantID"
      type="text"
      placeholder="Enter the restaurant ID"
      required
      autofocus
    />
  </div>
</form>
```

Restaurant ID

b.

```
<div class="form-field">
  <label for="restaurantName">Restaurant name</label>
  <input
    id="restaurantName"
    name="restaurantName"
    type="text"
    placeholder="Enter the restaurant name"
    required
  />
</div>
```

Restaurant name

c.

```
<div class="form-field">
  <label for="Email">Email</label>
  <input
    id="Email"
    name="Email"
    type="email"
    placeholder="Enter your email address"
    required
  />
</div>
```

Email

d.

```
<div class="form-field">
  <label for="Comments">Comments</label>
  <textarea
    id="Comments"
    name="Comments"
    rows="6"
    placeholder="Describe the inspection observations and findings"
    required
  ></textarea>
</div>
```

Comments

Describe the inspection observations and findings

e.

```
<div class="form-field">
  <label for="Result">Result</label>
  <select id="Result" name="Result" required>
    <option value="" selected disabled>Select an inspection result</option>
    <option value="Excellent Pass">Excellent Pass</option>
    <option value="Pass">Pass</option>
    <option value="Needs Reinspection">Needs Reinspection</option>
    <option value="Fail">Fail</option>
  </select>
</div>
```

Result

Select an inspection result

Result

✓ Select an inspection result

Excellent Pass

Pass

Needs Reinspection

Fail

Submit Inspection Record

f.

```
<div class="terms-row">
  <input id="termsAccepted" name="termsAccepted" type="checkbox" />
  <label for="termsAccepted">I agree to the terms and conditions.</label>
</div>
```

☐ I agree to the terms and conditions.

g.

```
<button type="submit">Submit Inspection Record</button>
```

Submit Inspection Record

5.

```
<script src="app.js"></script>
```

Part 1 – II Javascript

1.

a.

```
const validateForm = () => {  
  if (commentsInput.value.trim().length <= 25) {  
    alert("Comments must contain more than 25 characters.");  
    commentsInput.focus();  
    return false;  
  }  
}
```

This page says

Comments must contain more than 25 characters.

OK

aaa

Email

aaa@gmail.com

Comments

Test

b.

```
if (!termsCheckbox.checked) {  
  alert("You must agree to the terms and conditions before submitting.");  
  termsCheckbox.focus();  
  return false;  
}  
  
return true;  
};
```

This page says

Comments must contain more than 25 characters.

OK

aaa

Email

aaa@gmail.com

Comments

Test

Result

Excellent Pass

☐ I agree to the terms and conditions.

2.

```
const formData = new FormData(inspectionForm);
const formObject = {
  ...Object.fromEntries(formData.entries()),
  termsAccepted: termsCheckbox.checked,
};

const jsonString = JSON.stringify(formObject);
console.log("Form data JSON:", jsonString);
```

Form data JSON:
{ "restaurantID": "SF0001", "restaurantName": "Torihide", "Email": "eric.e.zhao@sjsu.edu", "Comments": "Every aspect of this restaurant is well above compliance requirements.", "Result": "Excellent Pass", "termsAccepted": true }

3.

```
const parsedObject = JSON.parse(jsonString);
const { restaurantID, Email } = parsedObject;
console.log("Restaurant ID:", restaurantID);
console.log("Email:", Email);
```

Restaurant ID: SF0001

Email: eric.e.zhao@sjsu.edu

4.

```
const updatedObject = {
  ...parsedObject,
  submissionDate: new Date().toISOString(),
};
console.log("Updated submission:", updatedObject);
```

Updated submission: **Object** {
 Comments: "Every aspect of this restaurant
 Email: "eric.e.zhao@sjsu.edu"
 Result: "Excellent Pass"
 restaurantID: "SF0001"
 restaurantName: "Torihide"
 submissionDate: "2026-08-30T23:06:48.432Z"
 termsAccepted: true

5.

```
const submissionCount = trackSuccessfulSubmission();
console.log("Successful submission count:", submissionCount);
```

☐ I agree to the terms and conditions.

Submit Inspection Record

Inspection submitted successfully. Submission count: 1

Successful submission count: 1

Part 1 – Deployment

Local Test:

```
Dockerfile x
Dockerfile

1 FROM nginx:alpine
2 RUN rm -rf /usr/share/nginx/html/*
3
4 COPY ["HW1-{{Eric Zhao}}.html", "/usr/share/nginx/html/index.html"]
5 COPY styles.css /usr/share/nginx/html/styles.css
6 COPY app.js /usr/share/nginx/html/app.js
7
8 EXPOSE 8521
9
10 CMD ["nginx", "-g", "daemon off;"]
```

Search

Only show running containers

		Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Memory (%)	Dis
		data260-hw1-contain	0342212eb70f	data260-hw1	8521:80	0%	13.12MB / 5.78G	0.22%	8.6

localhost:8521

Local Restaurant Inspection

Restaurant ID

Enter the restaurant ID

Restaurant name

Enter the restaurant name

AWS ECS:

Local Restaurant Inspection

Restaurant ID

Restaurant name

Email

Comments

Result

☐ I agree to the terms and conditions.

Submit Inspection Record

aws

Search

[Option+S]

Amazon Elastic Container Service

>

...

>

Tasks

>

520f5b0f7b1841c0b5d638f0e2037acd

>

Cor

Amazon Elastic Container Service

<

Favorites and recents

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions

Account settings

AWS Resource Explorer

AWS Batch

Amazon ECR

Repositories

Online learning workshop

Documentation

Discover products

Subscriptions

Configuration

Operating system/Architecture

Linux/ARM64

Capacity provider

Fargate

Task role

-

Task group

service:data260-hm1-service

[View service](#)

Fargate ephemeral storage

Encryption

Info

Default AWS Fargate encryption

Size (GiB)

20

Networking

Network mode

awsipc

ENI ID

[eni-0d948ff938a66d11e](#)

Private IP

[172.31.26.169](#)

Public IP

[54.215.206.119](#) | [open address](#)

Part 2 – Agentic AI

Command used:

```
(data260-hw1) eric@Eric-Mac hw1 % python agents_demo.py \  
--model qwen3:8b \  
--temperature 0.2 \  
--strict \  
--title "Local Restaurant Inspections" \  
--content "Observed multiple live flies landing on clean food-contact storage racks and food prep counters; small rodent droppings noted near the dry storage shelving baseboards."
```

Planner output:

```
--- Planner (15758 ms) ---  
{  
  "thought": "Identifying key issues in the inspection report to create relevant tags and a concise summary.",  
  "message": "Blog post highlights critical hygiene and pest control issues found during a local restaurant inspection.",  
  "data": {  
    "tags": [  
      "local restaurant inspections",  
      "local restaurant",  
      "restaurant inspections"  
    ],  
    "summary": "A recent inspection found pest infestation and hygiene issues in a local restaurant, raising food safety concerns.",  
    "issues": [  
      "Live flies on food-contact surfaces",  
      "Rodent droppings near storage areas",  
      "Potential health code violations"  
    ]  
  }  
}
```

Reviewer output:

```
--- Reviewer (26390 ms) ---  
{  
  "thought": "Validating tags for specificity and relevance to the inspection findings.",  
  "message": "Blog post highlights critical hygiene and pest control issues found during a local restaurant inspection.",  
  "data": {  
    "tags": [  
      "local restaurant inspections",  
      "local restaurant",  
      "restaurant inspections"  
    ],  
    "summary": "Inspection found flies on food surfaces and rodent droppings near storage, indicating hygiene and pest issues.",  
    "issues": [  
      "Live flies on food-contact surfaces",  
      "Rodent droppings near storage areas",  
      "Potential health code violations"  
    ]  
  }  
}
```

Finalized output:

```
Finalized Output  
{  
  "thought": "Finalizing tags and summary based on inspection findings and reviewer validation.",  
  "message": "Blog post highlights critical hygiene and pest control issues found during a local restaurant inspection.",  
  "data": {  
    "tags": [  
      "local restaurant inspections",  
      "restaurant hygiene",  
      "local restaurant"  
    ],  
    "summary": "Inspection found flies on food surfaces and rodent droppings near storage, indicating hygiene and pest issues.",  
    "issues": []  
  }  
}
```

Q1. The 3 final tags obtained.

“local restaurant inspections”, “restaurant hygiene”, “local restaurant”

Q2. Final Summary:

“Inspection found flies on food surfaces and rodent droppings near storage, indicating hygiene and pest issues.”

Q3. Yes. The Reviewer shortened and clarified the summary by focusing directly on the flies, rodent droppings, hygiene concerns, and pest issues.

The Planner creates the initial set of 3 relevant tags and a summary that is less than 25 words. The Reviewer received both the original input and the Planner's output. It checked whether the tags were relevant and specific, whether there were exactly three tags, and whether the summary was clear and within the word limit. In my run, the Reviewer replaced the general tag restaurant inspection with the more specific tag food safety inspection and shortened the summary. The Finalizer used the Planner's proposal and the Reviewer's feedback to prepare the final result. It kept the reviewed tags and summary, made sure the output followed the required structure, and cleared the issues list after finalization.

Part 3

Input

```
{
  "title": "Yakitori Restaurant Inspections",
  "content": "Observed multiple live flies landing on clean food-contact storage racks
and food prep counters; small rodent droppings noted near the dry storage shelving
baseboards."
}
```

JSON

```
{
  "model": "qwen3:8b",
  "input_file": "reports/hw01/cases/nondeterminism_input.json",
  "input_sha256": "8fe9960f63e8379b48eaf43a9f7a07c302f71e3200eb41d4a2fbba438a41761e",
  "result_count": 40,
  "results": [
    {
      "temperature": 0.7,
      "run": 1,
      "timestamp_utc": "2026-08-31T18:00:32Z",
      "latency_ms": 14209,
      "tags": [
        "food safety inspections",
        "restaurant hygiene",
        "yakitori restaurant inspections"
      ],
      "summary": "Inspection found live flies and rodent droppings in food storage
areas."
    },
    {
      "temperature": 0.7,
      "run": 2,
      "timestamp_utc": "2026-08-31T18:00:46Z",
      "latency_ms": 16794,
      "tags": [
        "yakitori restaurant inspections",
        "restaurant hygiene",
        "yakitori restaurant"
      ],
      "summary": "Yakitori restaurant inspection found live flies and rodent
droppings, indicating poor sanitation."
    },
    {
      "temperature": 0.7,
      "run": 3,
      "timestamp_utc": "2026-08-31T18:01:03Z",
```

```
    "latency_ms": 15427,
    "tags": [
      "restaurant hygiene",
      "yakitori inspections",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant inspection found flies and rodent droppings, showing hygiene and pest problems."
  },
  {
    "temperature": 0.7,
    "run": 4,
    "timestamp_utc": "2026-08-31T18:01:18Z",
    "latency_ms": 16889,
    "tags": [
      "food safety inspections",
      "yakitori restaurant inspections",
      "restaurant hygiene"
    ],
    "summary": "Yakitori inspection found live flies and rodent droppings, showing poor hygiene and pest control."
  },
  {
    "temperature": 0.7,
    "run": 5,
    "timestamp_utc": "2026-08-31T18:01:35Z",
    "latency_ms": 16420,
    "tags": [
      "yakitori restaurant inspections",
      "yakitori restaurant",
      "restaurant inspections"
    ],
    "summary": "A yakitori restaurant inspection found live flies and rodent droppings in storage areas."
  },
  {
    "temperature": 0.7,
    "run": 6,
    "timestamp_utc": "2026-08-31T18:01:52Z",
    "latency_ms": 16038,
    "tags": [
      "yakitori restaurant inspections",
      "yakitori restaurant",
      "restaurant inspections"
    ],
    "summary": "Yakitori restaurant inspection found live flies and rodent droppings, indicating poor hygiene and pest control."
  }
```

```
  },
  {
    "temperature": 0.7,
    "run": 7,
    "timestamp_utc": "2026-08-31T18:02:08Z",
    "latency_ms": 15036,
    "tags": [
      "food safety inspections",
      "yakitori restaurant inspections",
      "yakitori restaurant"
    ],
    "summary": "A yakitori restaurant had hygiene issues with flies and rodent droppings during an inspection."
  },
  {
    "temperature": 0.7,
    "run": 8,
    "timestamp_utc": "2026-08-31T18:02:23Z",
    "latency_ms": 14017,
    "tags": [
      "restaurant hygiene",
      "yakitori inspections",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene and pest control issues during an inspection."
  },
  {
    "temperature": 0.7,
    "run": 9,
    "timestamp_utc": "2026-08-31T18:02:37Z",
    "latency_ms": 16592,
    "tags": [
      "yakitori restaurant inspections",
      "food safety inspections",
      "yakitori restaurant"
    ],
    "summary": "Yakitori restaurant inspection found live flies and rodent droppings, indicating hygiene and food safety risks."
  },
  {
    "temperature": 0.7,
    "run": 10,
    "timestamp_utc": "2026-08-31T18:02:53Z",
    "latency_ms": 16236,
    "tags": [
      "food safety inspections",
```

```
    "restaurant hygiene",
    "yakitori restaurant inspections"
  ],
  "summary": "Inspection found live flies and rodent droppings in a yakitori restaurant, raising food safety concerns."
},
{
  "temperature": 0.7,
  "run": 11,
  "timestamp_utc": "2026-08-31T18:03:09Z",
  "latency_ms": 15260,
  "tags": [
    "yakitori restaurant inspections",
    "yakitori restaurant",
    "restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced live flies and rodent droppings, indicating hygiene failures."
},
{
  "temperature": 0.7,
  "run": 12,
  "timestamp_utc": "2026-08-31T18:03:25Z",
  "latency_ms": 14691,
  "tags": [
    "restaurant hygiene",
    "yakitori inspections",
    "yakitori restaurant inspections"
  ],
  "summary": "A yakitori restaurant had hygiene issues with flies and rodent droppings during an inspection."
},
{
  "temperature": 0.7,
  "run": 13,
  "timestamp_utc": "2026-08-31T18:03:39Z",
  "latency_ms": 15115,
  "tags": [
    "food safety inspections",
    "yakitori restaurant inspections",
    "yakitori restaurant"
  ],
  "summary": "A yakitori restaurant faced hygiene issues with live flies and rodent droppings during an inspection."
},
{
  "temperature": 0.7,
```

```
    "run": 14,
    "timestamp_utc": "2026-08-31T18:03:55Z",
    "latency_ms": 14594,
    "tags": [
      "yakitori restaurant inspections",
      "restaurant hygiene",
      "yakitori restaurant"
    ],
    "summary": "Inspection found live flies and rodent droppings in a yakitori restaurant."
  },
  {
    "temperature": 0.7,
    "run": 15,
    "timestamp_utc": "2026-08-31T18:04:09Z",
    "latency_ms": 15866,
    "tags": [
      "food safety inspections",
      "yakitori hygiene",
      "yakitori restaurant inspections"
    ],
    "summary": "Inspection found live flies and rodent droppings, indicating poor hygiene and pest control."
  },
  {
    "temperature": 0.7,
    "run": 16,
    "timestamp_utc": "2026-08-31T18:04:25Z",
    "latency_ms": 14296,
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant inspections",
      "yakitori restaurant"
    ],
    "summary": "Inspection found live flies and rodent droppings in a yakitori restaurant, raising food safety concerns."
  },
  {
    "temperature": 0.7,
    "run": 17,
    "timestamp_utc": "2026-08-31T18:04:39Z",
    "latency_ms": 16707,
    "tags": [
      "food safety inspections",
      "restaurant hygiene",
      "yakitori restaurant inspections"
    ],
  },
```

```
    "summary": "Inspection found live flies and rodent droppings in a yakitori restaurant, showing poor hygiene and pest control."
  },
  {
    "temperature": 0.7,
    "run": 18,
    "timestamp_utc": "2026-08-31T18:04:56Z",
    "latency_ms": 15574,
    "tags": [
      "restaurant hygiene",
      "food safety",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant had live flies and rodent droppings, showing poor hygiene and pest control."
  },
  {
    "temperature": 0.7,
    "run": 19,
    "timestamp_utc": "2026-08-31T18:05:12Z",
    "latency_ms": 16522,
    "tags": [
      "yakitori restaurant inspections",
      "yakitori restaurant",
      "restaurant inspections"
    ],
    "summary": "A yakitori restaurant faced hygiene issues with flies and rodent droppings in storage areas."
  },
  {
    "temperature": 0.7,
    "run": 20,
    "timestamp_utc": "2026-08-31T18:05:28Z",
    "latency_ms": 15973,
    "tags": [
      "yakitori restaurant inspections",
      "food safety inspections",
      "restaurant hygiene"
    ],
    "summary": "A yakitori restaurant inspection found live flies and rodent droppings, showing poor hygiene."
  },
  {
    "temperature": 0.0,
    "run": 1,
    "timestamp_utc": "2026-08-31T18:05:44Z",
    "latency_ms": 14484,
```



```
{
  "temperature": 0.0,
  "run": 5,
  "timestamp_utc": "2026-08-31T18:06:37Z",
  "latency_ms": 12634,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 6,
  "timestamp_utc": "2026-08-31T18:06:49Z",
  "latency_ms": 12618,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 7,
  "timestamp_utc": "2026-08-31T18:07:02Z",
  "latency_ms": 12687,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 8,
  "timestamp_utc": "2026-08-31T18:07:14Z",
  "latency_ms": 12611,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
```

```
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent
droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 9,
  "timestamp_utc": "2026-08-31T18:07:27Z",
  "latency_ms": 12676,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent
droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 10,
  "timestamp_utc": "2026-08-31T18:07:40Z",
  "latency_ms": 12649,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent
droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 11,
  "timestamp_utc": "2026-08-31T18:07:52Z",
  "latency_ms": 12614,
  "tags": [
    "restaurant hygiene",
    "yakitori restaurant",
    "yakitori restaurant inspections"
  ],
  "summary": "Yakitori restaurant faced hygiene issues with flies and rodent
droppings during inspection."
},
{
  "temperature": 0.0,
  "run": 12,
```



```
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  },
  {
    "temperature": 0.0,
    "run": 16,
    "timestamp_utc": "2026-08-31T18:08:56Z",
    "latency_ms": 12619,
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  },
  {
    "temperature": 0.0,
    "run": 17,
    "timestamp_utc": "2026-08-31T18:09:09Z",
    "latency_ms": 12682,
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  },
  {
    "temperature": 0.0,
    "run": 18,
    "timestamp_utc": "2026-08-31T18:09:21Z",
    "latency_ms": 12652,
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  },
  {
    "temperature": 0.0,
    "run": 19,
    "timestamp_utc": "2026-08-31T18:09:34Z",
    "latency_ms": 12665,
```

```
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  },
  {
    "temperature": 0.0,
    "run": 20,
    "timestamp_utc": "2026-08-31T18:09:47Z",
    "latency_ms": 12624,
    "tags": [
      "restaurant hygiene",
      "yakitori restaurant",
      "yakitori restaurant inspections"
    ],
    "summary": "Yakitori restaurant faced hygiene issues with flies and rodent droppings during inspection."
  }
]
```

Based on my results, two users submitting the identical input at temperature 0.7 might receive different combinations of tags. My 20 runs produced seven distinct tag sets. At temperature 0.0, all 20 runs produced the same tag set, so two users would be much more likely to receive identical tags.

Run-to-run variation is acceptable when generating descriptive tags for a restaurant-inspection article because several different relevant phrases can describe the same content. For example, food safety inspections and restaurant hygiene are both reasonable tags.

Variation would not be acceptable when deciding whether a restaurant passed an inspection or must close. A safety-critical decision should produce a consistent result when the inspection evidence is identical.

Part 4

Why is prior conversation context resent with every turn?

Each model request is stateless. To let the model respond as part of an ongoing conversation, the client sends the system prompt and earlier user/assistant messages again with the new user message.

How is a system prompt different from a user message?

A system prompt sets the model's overall role and behavior, such as the bullet-only code-review format. A user message supplies the current request. The system instruction has higher priority and applies across the conversation.

Why do input tokens grow over a conversation?

Every new request includes the prior conversation, so the input contains more text after each turn. Earlier messages may be counted repeatedly because they are sent again as context.

What eventually limits that growth?

The model has a finite context window. Once the system prompt, history, and new request approach that limit, the client must remove, summarize, or truncate older context. Increasing input size also increases latency and resource use before the hard limit is reached.

See github:

<https://github.com/ericezhao/data260-1421/tree/main>